

When Does Focal Loss Help? A Controlled Comparison of Focal Loss and Cross-Entropy for Small-CNN Image Classification on Balanced and Long-Tailed CIFAR-10

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Abstract—Focal loss was introduced to address foreground-background imbalance in dense object detection, yet it is often adopted as a generic drop-in replacement for cross-entropy in image classification. We ask a precise question: does focal loss beat cross-entropy for CIFAR-10 classification with a small convolutional neural network? Holding a 0.31M-parameter CNN, optimizer, schedule, and data augmentation fixed, we vary only the loss function (cross-entropy versus focal loss with $\gamma \in \{1, 2, 3\}$) and the class distribution (balanced CIFAR-10 versus long-tailed CIFAR-10-LT with imbalance factor 100), and we report top-1 accuracy, Expected Calibration Error (ECE), macro-F1, and worst-class accuracy. On balanced CIFAR-10, cross-entropy attains the best accuracy (87.90%); focal loss never exceeds it and degrades monotonically with γ , its only benefit being improved calibration at $\gamma = 1$ (ECE 0.0154 versus 0.0337). Under class imbalance the conclusion reverses: focal loss with $\gamma = 3$ improves accuracy (59.10% versus 57.66%), macro-F1 (by 3.2 points), worst-class accuracy (3.6% to 13.3%, a $3.7\times$ gain), and calibration (ECE 0.220 to 0.050). We conclude that focal loss is a tool for imbalance and calibration, not a generic accuracy upgrade, and we provide practical guidance on selecting γ . All code and results are released.

Index Terms—focal loss, cross-entropy, image classification, class imbalance, model calibration, CIFAR-10

I. INTRODUCTION

The choice of loss function is among the most consequential decisions when training a classifier. Softmax cross-entropy is the default for image classification, but a steady stream of alternatives promises improvements on specific failure modes. Focal loss [1] is one of the most widely adopted: by multiplying the cross-entropy term with a modulating factor that downweights well-classified examples, it was designed to combat the extreme foreground-background imbalance encountered in one-stage dense object detection.

Because of its success in detection, focal loss is frequently reused as a generic replacement for cross-entropy in classification tasks, often without verifying that the conditions that made it effective are present. This motivates a precise and falsifiable question: *does focal loss beat cross-entropy for CIFAR-10 classification with a small CNN?* CIFAR-10 is a balanced benchmark, so a priori there is little of the easy-example dominance that focal loss is designed to suppress.

We answer this question with a tightly controlled study. We fix a small convolutional network, its optimizer, learning-

rate schedule, and data augmentation, and vary only two factors: the loss function (cross-entropy versus focal loss with $\gamma \in \{1, 2, 3\}$) and the class distribution (the standard balanced CIFAR-10 versus a long-tailed variant, CIFAR-10-LT, with imbalance factor 100). Figure 1 summarizes the design. We evaluate four complementary metrics: top-1 accuracy, Expected Calibration Error (ECE), macro-F1, and worst-class accuracy.

Contributions.

- A clean, fully reproducible head-to-head comparison of cross-entropy and focal loss ($\gamma \in \{1, 2, 3\}$) on balanced and long-tailed CIFAR-10 with a single fixed small CNN.
- Evidence that on balanced data cross-entropy wins on accuracy and focal loss helps only calibration, and only at low γ .
- Evidence that under class imbalance the verdict reverses on every metric, with larger γ preferred, and a $3.7\times$ improvement in worst-class accuracy.
- Practical guidance on when to use focal loss, with open code and results executed on serverless GPUs.

The remainder of the paper reviews related work (Section II), formalizes the losses and metrics (Section III), details the experimental setup (Section IV), presents results (Section V), and discusses limitations (Section VI) and conclusions (Section VII).

II. BACKGROUND AND RELATED WORK

A. Focal Loss and Class Imbalance

Focal loss [1] reshapes cross-entropy to focus training on hard examples and was central to the RetinaNet detector. In long-tailed recognition, reweighting and margin-based methods are the dominant tools: the Class-Balanced loss [3] reweights by the effective number of samples and, in its CB-Focal form, improves on plain focal loss under imbalance; LDAM [4] enforces label-distribution-aware margins. These works establish focal loss as a baseline for imbalance rather than a general-purpose accuracy method.

B. Calibration of Neural Networks

Modern networks are often miscalibrated, tending to be overconfident [5]. Mukhoti et al. [2] show that focal loss

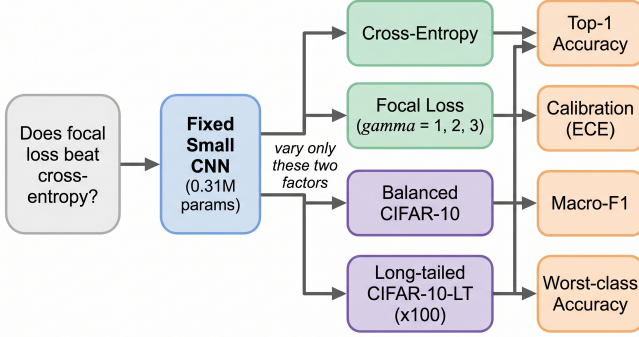


Fig. 1. Controlled experimental design. Only the loss function and the class distribution are varied; the model, optimizer, schedule, and augmentation are held fixed. Each configuration is evaluated on four metrics.

produces substantially better calibrated classifiers than cross-entropy without sacrificing accuracy, and that combining it with temperature scaling yields state-of-the-art calibration. Their analysis predicts a calibration benefit even on balanced data, which our experiments test directly.

C. Long-Tailed Recognition Benchmarks

Long-tailed CIFAR variants, constructed by exponentially subsampling classes, are a standard testbed for imbalance [3], [6]. We adopt the same construction with an imbalance factor of 100.

III. METHODOLOGY

A. Problem Formulation

For an input with ground-truth class y and predicted class probabilities p , let $p_t = p_y$ denote the probability assigned to the true class. Softmax cross-entropy is

$$\mathcal{L}_{\text{CE}} = -\log(p_t). \quad (1)$$

Focal loss [1] introduces a modulating factor with focusing parameter $\gamma \geq 0$:

$$\mathcal{L}_{\text{FL}} = -(1 - p_t)^\gamma \log(p_t). \quad (2)$$

When $\gamma = 0$, focal loss reduces exactly to cross-entropy; as γ increases, the loss contributed by easy, well-classified examples ($p_t \rightarrow 1$) is increasingly suppressed. We use the multi-class form with $\gamma \in \{1, 2, 3\}$ and probabilities clamped to $[10^{-7}, 1]$ for numerical stability.

B. Small CNN Architecture

We use a compact convolutional network (Fig. 2) of three convolutional blocks with 32, 64, and 128 channels. Each block applies two Conv-BatchNorm-ReLU units followed by 2×2 max pooling. A global average pool feeds a fully connected layer of width 128 with dropout 0.3 and a final linear classifier over the ten classes. The network has 305,706 trainable parameters. The hyperparameters are listed in Table I.

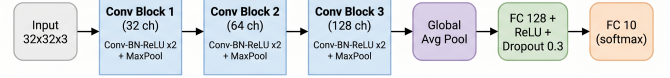


Fig. 2. The fixed 0.31M-parameter small CNN used in all experiments: three convolutional blocks followed by global average pooling and a compact classifier head.

TABLE I
ARCHITECTURE AND TRAINING HYPERPARAMETERS

Setting	Value
Parameters	305,706
Optimizer	AdamW (lr 10^{-3} , weight decay 5×10^{-4})
Schedule	Cosine annealing
Epochs	30
Batch size	128
Augmentation	RandomCrop(32, pad 4), horizontal flip
Dropout	0.3
Hardware	Modal serverless NVIDIA A100-40GB
Framework	PyTorch 2.4

C. Long-Tailed Data Construction

To probe the imbalanced regime we build CIFAR-10-LT by subsampling the training set so that the number of examples per class follows an exponential profile. For class index $c \in \{0, \dots, 9\}$ and imbalance factor ρ ,

$$n_c = n_{\max} \rho^{-c/(C-1)}, \quad (3)$$

with $n_{\max} = 5000$, $C = 10$, and $\rho = 100$, giving per-class counts $[5000, 2997, 1797, 1077, 646, 387, 232, 139, 83, 50]$. The test set remains balanced.

D. Evaluation Metrics

We report top-1 accuracy; macro-F1 (the unweighted mean of per-class F1, which exposes tail performance); worst-class accuracy (the minimum per-class accuracy); and the Expected Calibration Error with $M = 15$ equal-width confidence bins,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (4)$$

where B_m is the set of predictions whose confidence falls in bin m .

IV. EXPERIMENTAL SETUP

A. Datasets

We use the standard CIFAR-10 dataset (50,000 training and 10,000 test images, 32×32 RGB, ten balanced classes)

TABLE II
BALANCED CIFAR-10 RESULTS (SMALL CNN, 30 EPOCHS)

Loss	Acc	ECE	Macro-F1	Worst-cls
Cross-entropy	0.8790	0.0337	0.8786	0.729
Focal $\gamma = 1$	0.8744	0.0154	0.8741	0.743
Focal $\gamma = 2$	0.8728	0.0660	0.8724	0.715
Focal $\gamma = 3$	0.8638	0.1096	0.8632	0.692

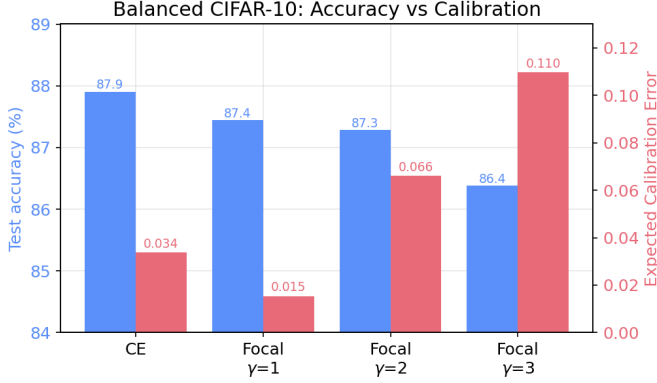


Fig. 3. On balanced CIFAR-10, cross-entropy yields the best accuracy while focal loss with $\gamma = 1$ yields the best calibration; larger γ degrades both accuracy and calibration.

and the long-tailed CIFAR-10-LT variant described above (approximately 12,408 training images, balanced test set).

B. Implementation Details

All runs use the architecture and hyperparameters of Table I, executed on Modal serverless A100-40GB GPUs with a fixed random seed. Each run trains for 30 epochs in roughly two to four minutes; the full study used about twenty minutes of GPU time.

C. Compared Configurations

On balanced CIFAR-10 we compare cross-entropy against focal loss with $\gamma \in \{1, 2, 3\}$. On CIFAR-10-LT we compare cross-entropy against focal loss with $\gamma \in \{2, 3\}$, the settings expected to be most effective under imbalance.

V. RESULTS

A. Balanced CIFAR-10

Table II and Fig. 3 report the balanced results. Cross-entropy achieves the best top-1 accuracy (87.90%). Focal loss does not exceed it at any γ , and accuracy decreases monotonically as γ grows (87.44%, 87.28%, 86.38%). The one benefit of focal loss appears in calibration: at $\gamma = 1$ it more than halves the ECE relative to cross-entropy (0.0154 versus 0.0337). However, at $\gamma \geq 2$ the calibration error exceeds that of cross-entropy, as the strong down-weighting makes the model underconfident. On balanced data, therefore, the only advantage of focal loss is improved calibration, and only at low γ .

TABLE III
LONG-TAILED CIFAR-10-LT RESULTS (IMBALANCE FACTOR 100)

Loss	Acc	ECE	Macro-F1	Worst-cls
Cross-entropy	0.5766	0.2202	0.5332	0.036
Focal $\gamma = 2$	0.5908	0.0967	0.5618	0.114
Focal $\gamma = 3$	0.5910	0.0503	0.5655	0.133

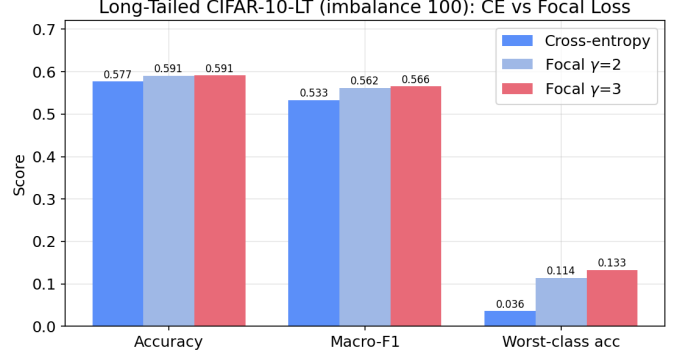


Fig. 4. Under imbalance (factor 100), focal loss improves accuracy, macro-F1, and especially worst-class accuracy over cross-entropy.

B. Long-Tailed CIFAR-10-LT

Table III and Fig. 4 report the imbalanced results. Here the verdict reverses: focal loss improves on cross-entropy across all metrics, and larger γ helps more. At $\gamma = 3$, accuracy rises from 57.66% to 59.10%, macro-F1 from 0.5332 to 0.5655, and ECE falls from 0.2202 to 0.0503. The most striking effect is on the rarest classes: worst-class accuracy increases from 3.6% to 13.3%, a $3.7\times$ improvement. Figure 5 plots per-class accuracy ordered by training frequency and shows that the gains are concentrated in the tail, where cross-entropy collapses while focal loss preserves a meaningful fraction of accuracy. Figure 6 shows the training dynamics.

C. Discussion

The two regimes tell a coherent story. On balanced data, every class already contributes a similar share of easy and hard examples, so the modulating factor of focal loss mostly removes useful gradient signal, slightly reducing accuracy; only mild focusing ($\gamma = 1$) helps, and solely by tempering overconfidence. Under heavy imbalance, the abundant head classes dominate the cross-entropy gradient and the tail is effectively ignored, which is exactly the easy-example dominance focal loss was built to counter. Down-weighting confident head predictions lets the rare classes contribute, so larger γ yields better accuracy, macro-F1, calibration, and in particular worst-class accuracy. This matches the original motivation of focal loss [1] and the calibration findings of Mukhoti et al. [2].

VI. LIMITATIONS AND FUTURE WORK

Our study fixes one architecture, one dataset family, one imbalance factor, and a single seed per configuration, so

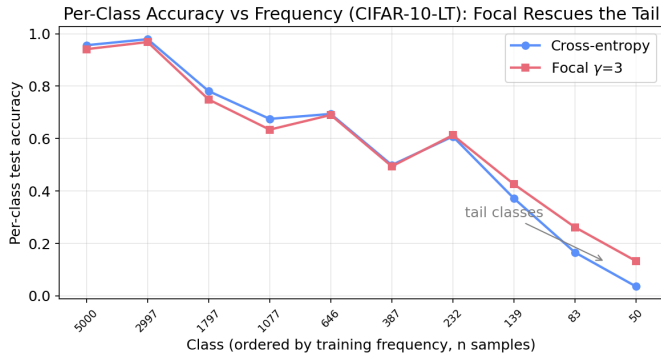


Fig. 5. Per-class accuracy ordered by training frequency on CIFAR-10-LT. Focal loss ($\gamma = 3$) recovers accuracy on the rarest classes (right) where cross-entropy collapses.

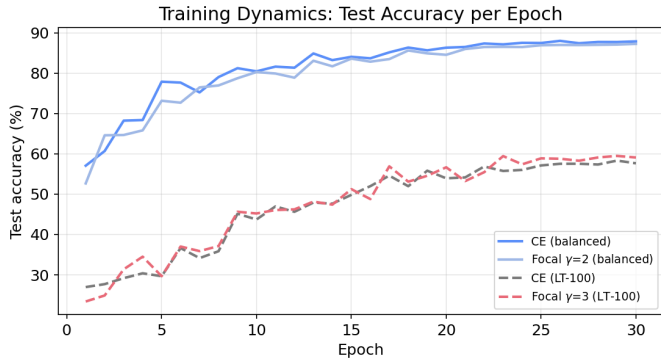


Fig. 6. Training dynamics: test accuracy per epoch for representative configurations on both the balanced and long-tailed settings.

we report point estimates rather than confidence intervals. We do not apply temperature scaling, which is known to further improve calibration, nor do we combine focal loss with explicit reweighting such as CB-Focal [3]. Future work should add multiple seeds with error bars, larger backbones (e.g., ResNet), additional datasets including ImageNet-LT, a sweep over imbalance factors, and a comparison against dedicated long-tailed methods.

VII. CONCLUSION

We asked whether focal loss beats cross-entropy for CIFAR-10 classification with a small CNN. The answer is no on the standard balanced benchmark: cross-entropy gives the best accuracy and focal loss helps only calibration, and only at low γ . The answer becomes yes under class imbalance, where focal loss improves every metric and dramatically rescues rare-class accuracy, with larger γ preferred. Focal loss is therefore best understood as a targeted remedy for imbalance and miscalibration rather than a generic replacement for cross-entropy. Practitioners working with balanced data should default to cross-entropy, while those facing long-tailed

distributions should consider focal loss with a relatively large focusing parameter.

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